Practice workbook · Series made of logarithms

Series where terms are logarithms (or are turned into logarithms), so log laws link a geometric pattern to an arithmetic one.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-15

1. Question 4 · part whole · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part whole only.

4 The nth term of an arithmetic series is u_n where

$$u_n = (n + 1)\ln 4$$

Given that the sum of the first n terms of the series is S_n

show that $S_n = \ln 2^{(n^2+an)}$ where a is an integer whose value is to be found.

(5)

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1. Question 4 · part whole · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 5 marks)

1. Question 4 · part whole · 5 marks · Working and answer

Working and answer

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2. Question 3 · part whole · 7 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part whole only.

3 The nth term of an arithmetic series is u_n such that

$$u_n = \ln a + (n - 1)\ln b$$

where a and b are positive integers.

Given that $u_2 = \ln 12$ and that $u_5 = \ln 768$

find the value of a and the value of b.

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2. Question 3 · part whole · 7 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 7 marks)

2. Question 3 · part whole · 7 marks · Working and answer

Working and answer

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3. Question 5 · part c · 5 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

5 A geometric series G has first term 12 and common ratio $\frac{3}{8}$

(a) Find the sum to infinity of G

(2)

(b) Show that the 6th term of G can be written as $\frac{3^6}{2^{13}}$

(3)

The nth term of G is u_n

(c) By finding an expression for u_n in terms of n, show that

$$\log_2 u_n = n\log_2 3 - 3n + 5$$

(5)

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3. Question 5 · part c · 5 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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3. Question 5 · part c · 5 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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3. Question 5 · part c · 5 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 10 marks)

3. Question 5 · part c · 5 marks · Working and answer

Working and answer

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